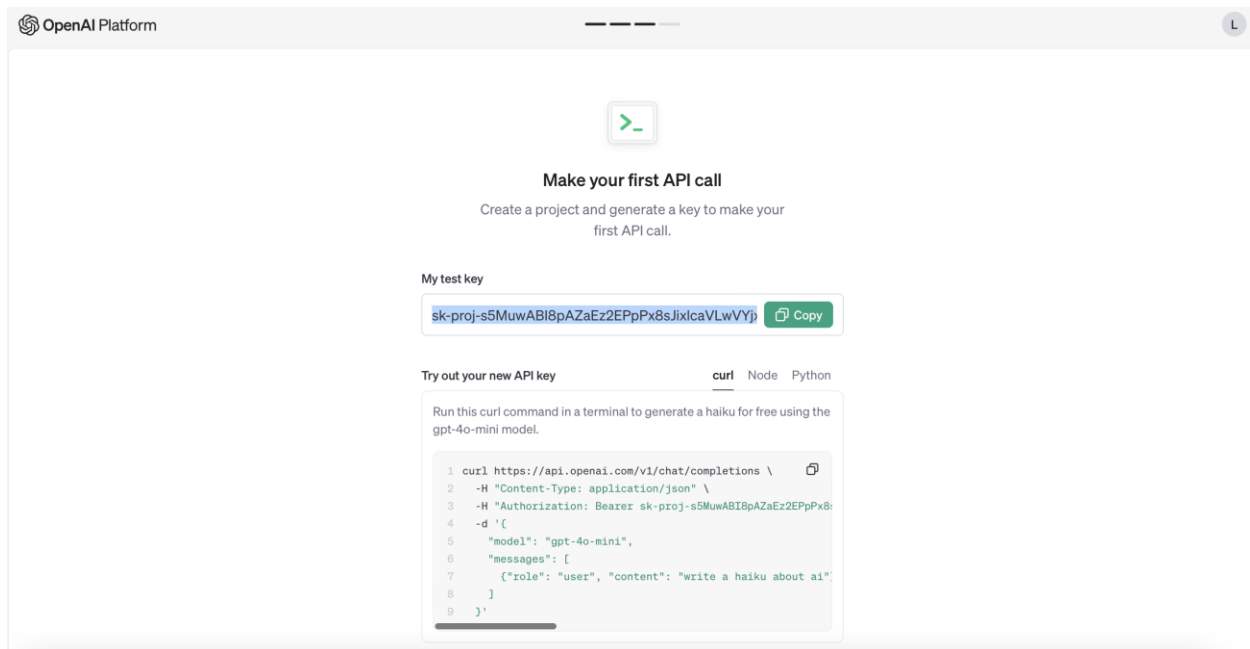


AI Coding

Task #13: Setting Up Cursor (A coding environment)

Step 1:

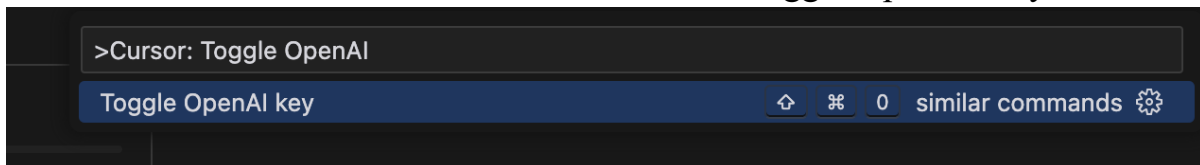
I logged into the OpenAI Platform and generated a new API key.



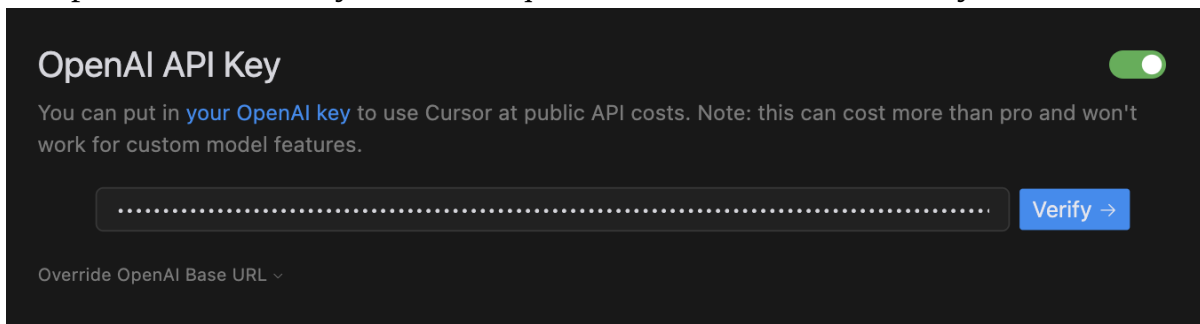
The API key is required to connect Cursor with the OpenAI model.

Step 2:

I opened Cursor and searched for the command “Cursor: Toggle OpenAI Key” and enabled it.



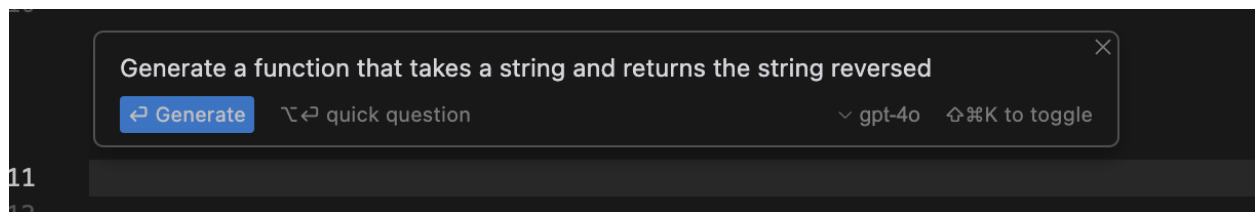
After that, I pasted the API key into the required field and clicked Verify.



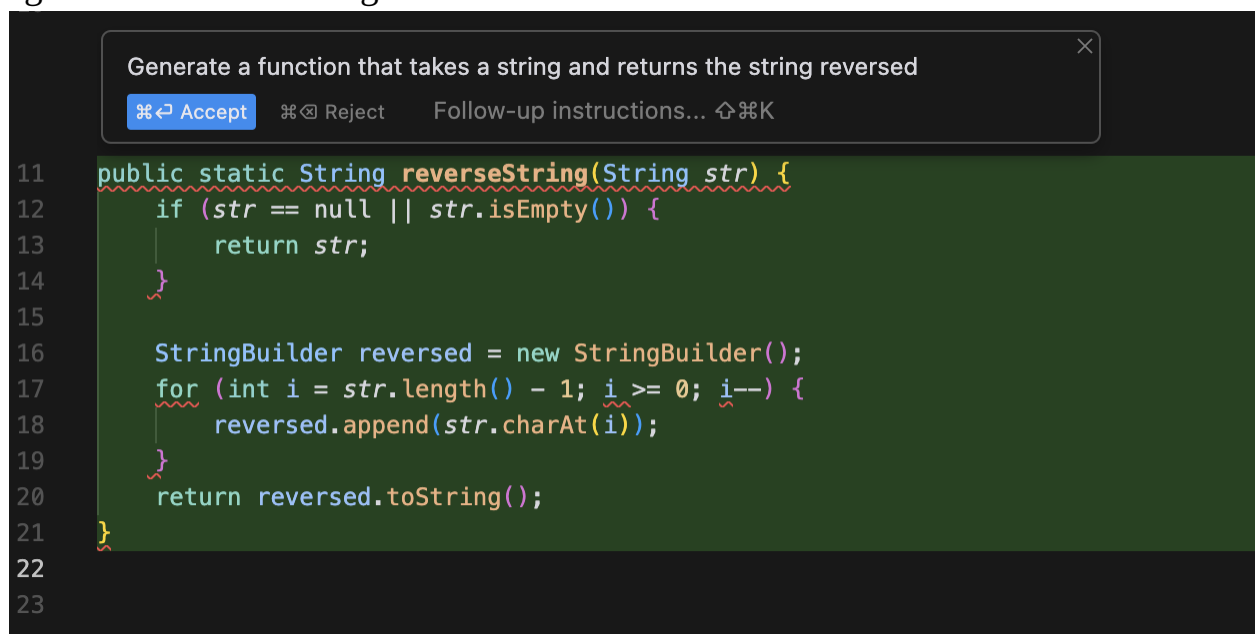
| ⌘L to chat, ⌘K to generate

Task #15: Iterative coding in Cursor

I started by asking Cursor to generate a simple function:

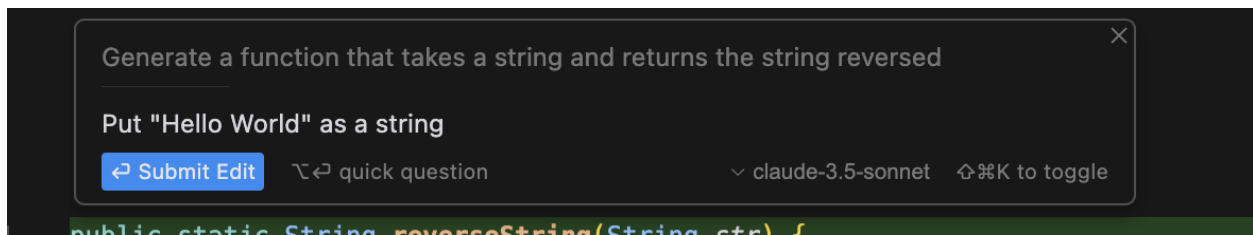


Cursor generated the following function:

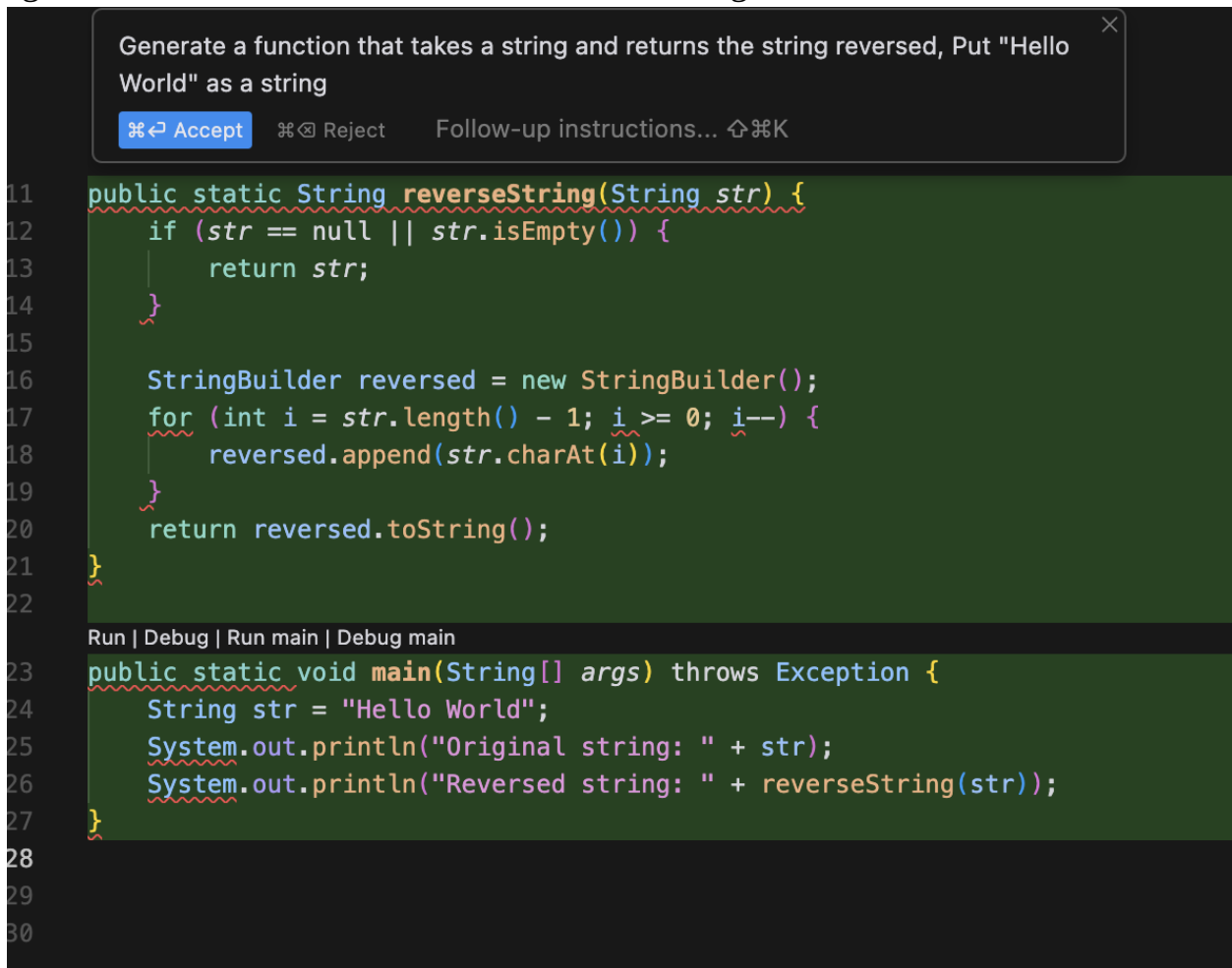


The AI assistant quickly provided a function to reverse a string using a `StringBuilder` and a `for` loop.

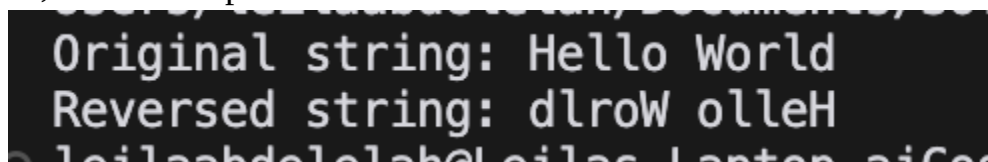
Next, I wanted to test the function with a sample input. I provided another instruction:



Cursor generated a main method to test the reverseString function:



I ran the program, and the output was:



I decided to extend the functionality further by asking the AI to help me count the letters in a string. I provided the following prompt to Cursor. Cursor generated the following function to count each letter's occurrences:

```
public class App {  
    Add a letter counter function in this class  
    Accept Reject Follow-up instructions... K  
  
    public static String letterCounter(String str) {  
        if (str == null || str.isEmpty()) {  
            return "Empty string";  
        }  
  
        int[] counts = new int[26];  
        StringBuilder result = new StringBuilder();  
  
        // Count each letter  
        for (char c : str.toLowerCase().toCharArray()) {  
            if (Character.isLetter(c)) {  
                counts[c - 'a']++;  
            }  
        }  
  
        // Build result string  
        for (int i = 0; i < 26; i++) {  
            if (counts[i] > 0) {  
                result.append((char)('a' + i))  
                    .append(str: ": ")  
                    .append(counts[i])  
                    .append(str: "\n");  
            }  
        }  
  
        return result.length() > 0 ? result.toString() : "No letters found";  
    }  
}
```

To test this function, I updated the main method and added the following line:

```
Run | Debug | Run main | Debug main  
public static void main(String[] args) throws Exception {  
    String str = "Hello World";  
    System.out.println("Original string: " + str);  
    System.out.println("Reversed string: " + reverseString(str));  
    System.out.println("Letter counts: " + letterCounter(str));  
}  
}
```

This is the output that I got:

```
Users/teitaaudeletan/Documents/
Original string: Hello World
Reversed string: dlrow olleH
Letter counts: d: 1
e: 1
h: 1
l: 3
o: 2
r: 1
w: 1
```

Next, I wanted the function to count the total number of letters instead of showing individual letter counts. I prompted the AI again:

```
This function should count total number of letters in the sentence
Submit Edit quick question claude-3.5-sonnet K to toggle

public static String letterCounter(String str) {
    if (str == null || str.isEmpty()) {
        return "Empty string";
    }

    int[] counts = new int[26];
    StringBuilder result = new StringBuilder();

    // Count each letter
    for (char c : str.toLowerCase().toCharArray()) {
        if (Character.isLetter(c)) {
            counts[c - 'a']++;
        }
    }
}
```

The function was modified to include a letterCount variable that keeps track of the total number of letters:

This function should count total number of letters in the sentence

Accept

Reject

Follow-up instructions... K

```
public static String letterCounter(String str) {
    if (str == null || str.isEmpty()) {
        return "Empty string";
    }

    int[] counts = new int[26];
    StringBuilder result = new StringBuilder();
    int letterCount = 0;

    // Count each letter
    for (char c : str.toLowerCase().toCharArray()) {
        // Count total letters
        for (char c : str.toCharArray()) {
            if (Character.isLetter(c)) {
                counts[c - 'a']++;
                letterCount++;
            }
        }
    }

    // Build result string
    for (int i = 0; i < 26; i++) {
        if (counts[i] > 0) {
            result.append((char)('a' + i))
                .append(": ")
                .append(counts[i])
                .append("\n");
        }
    }

    return result.length() > 0 ? result.toString() : "No letters found";
    return letterCount > 0 ? "Total letters: " + letterCount : "No letters found";
}
```

```
3
4 public static String letterCounter(String str) {
5     if (str == null || str.isEmpty()) {
6         return "Empty string";
7     }
8
9     int letterCount = 0;
10
11     // Count total letters
12     for (char c : str.toCharArray()) {
13         if (Character.isLetter(c)) {
14             letterCount++;
15         }
16     }
17
18     return letterCount > 0 ? "Total letters: " + letterCount : "No letters found";
19 }
```

I ran the updated program, and the output now included the total count of letters:

```
Original string: Hello World  
Letter counts: Total letters: 10  
Reversed string: dlrow olleH
```

Experience:

- Setting up Cursor was straightforward once I had the OpenAI API key.
- The AI assistant was very helpful in generating the function quickly without errors.
- I liked how Cursor could also provide test cases and integrate them into the code seamlessly.
- Cursor quickly added the letter counter function and iteratively improved it as I refined my requirements.
- I liked how the AI could integrate the new logic seamlessly into the existing class.
- Testing was easy because Cursor helped with adding lines to the main method for debugging.

Challenges:

- At first, I didn't know where to enter the API key, but I found the "Toggle OpenAI Key" option in the Command Palette.
- I needed to phrase my prompts clearly
- Initially, the function only displayed letter counts without the total count, but by clarifying my prompt, I was able to achieve the desired output.